Development of Affordable and Portable Atom-Scanning Technology

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Our goal

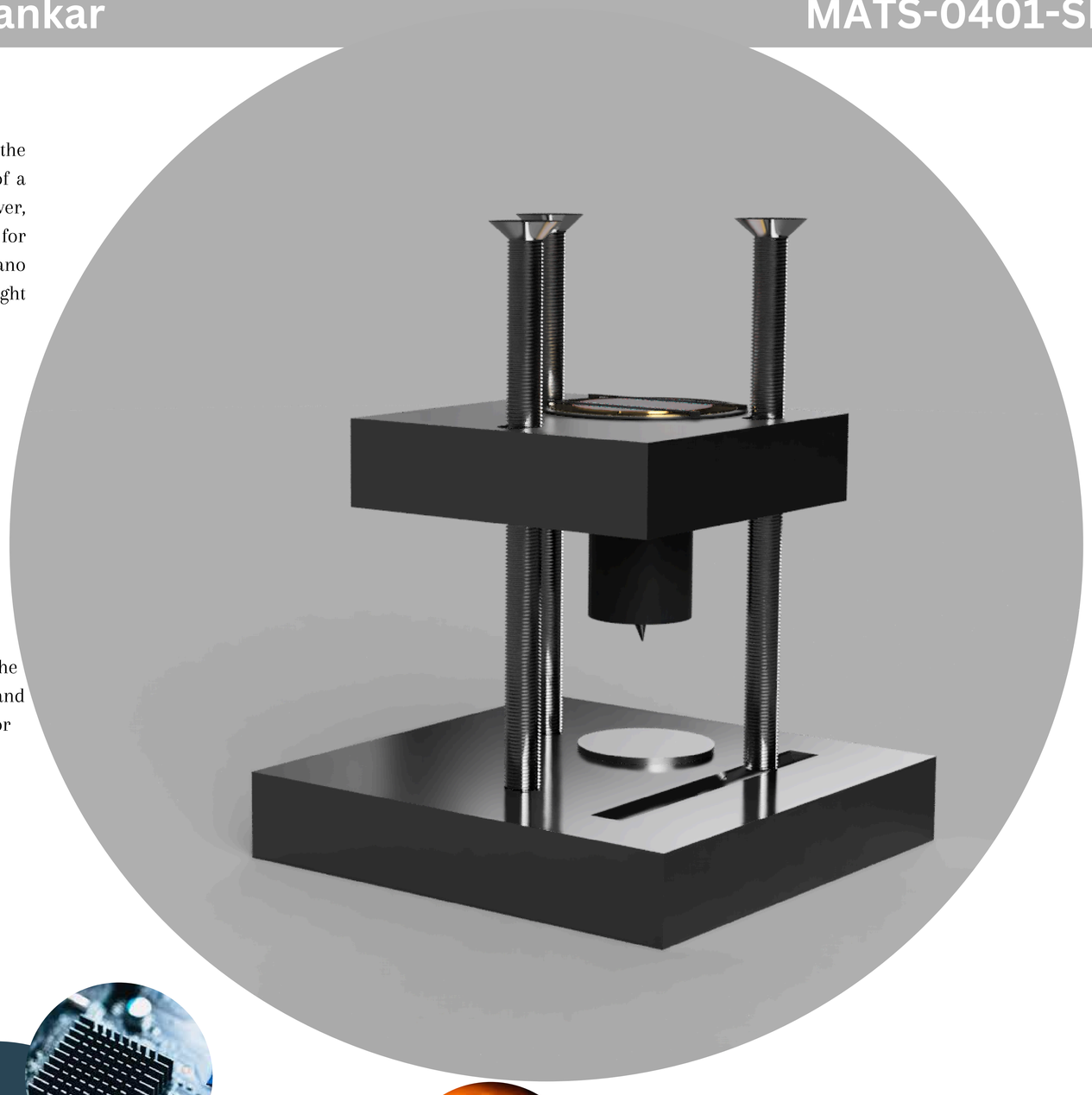
Modern technology is reaching deeper and deeper into the depths of the atomic world. These innovations are possible thanks to the development of a novel technology called The Scanning Tunneling Microscope (STM). However, the common STM is too expensive and bulky, making it difficult to access for independent researchers and high school students. So our Portable Nano Scanning Technology recreates this expensive machinery with cheap and light parts with a novel design that makes the STM affordable for all.

Objective

Develop an cheap and portable model of the STM

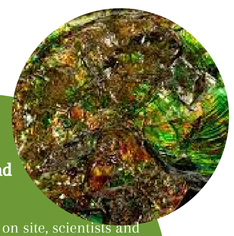
Our Impact

Our model of the PNST was built with other people in mind. The low-cost of our PNST was reached by keeping high-schoolers and individual researchers in mind who may not have the ability or the access to a proper STM. The portable and cheap design of our model finds its purpose in various fields.



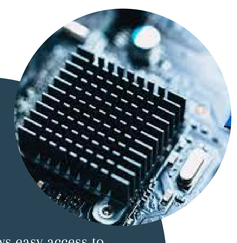
Increasing accessibility to high schoolers

The PNST can be easily rebuilt by other small high school research groups for less than $60. This allows these kids to play around with nano technology and explore the atomic world without having to reach out to different labs, and also possibly incorporate it into the class curriculum



On site analysis and studying

Often times while working on site, scientists and geologists come across materials whose properties can't be studied as closely due to the unavailability of a conventional STM. However thanks to our portable design, the scientists can get a rough idea of the properties before conducting further analysis.



Quality control

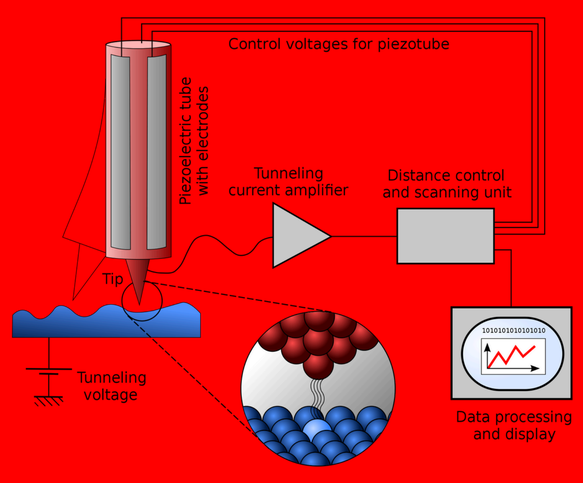
The low cost of the PNST allows easy access to inspectors for quality control of industry products like electrical components, nano materials, and products which require high precision.

Studying extreme environments

The portability of the model allows it to be incorporated into robots to study materials located deep in volcanos where humans can't travel. This allows for the creation of heavy duty robots that can remain in these environments and study them, allowing for deeper investigation of uknown territory

Background

- The STM is a pivotal sector of research in viewing microscopic scale structures, primarily due to its ability to provide atomic-scale imaging
- The sections of the STM are as followed:
  - Atomically sharp tip, a conductive tip often made of tungsten, sharpened to a single atom at the end to provide a high resolution for scanning the surface.
  - Piezoelectric scanner, a device precisely moving the tip in three dimensions (x, y, and z-axes) with atomic accuracy using the piezoelectric effect which is the expansion or contraction of materials when an electric field is applied.
  - Current amplifier, amplifying a miniscule tunneling current passing the vacuum gap from the sample to the tip, and then using the relative voltage difference to map out topography of the sample.
  - Feedback loop adjusts the tip's height to maintain a constant tunneling current enabling the mapping of the surfaces topography.
- The science behind STM's ties closely with several key quantum mechanical phenomena, mainly, quantum tunneling. To explain this, we first define a few key terms
  - Fermi energy is the highest energy electrons occupy at absolute zero temperature
  - Bias voltage establishes an energy differential to aid in the tunneling process by modulating the tunneling current by influencing the energy difference between the vacuum and the Fermi energy.
  - When we apply this voltage to an electron, it receives enough energy to "tunnel" but not cross the energy difference, which is known as quantum tunneling.
- The STM's functionality is governed by its atomically sharp tip. The functionality of the tip within the STM is to facilitate the localized and intense tunneling current that is necessary to produce high-resolution images. When quantum tunneling occurs, electrons from the surface of the sample are transferred to the tip, and captured carefully by the current amplifier. This phenomenon allows the STM to construct images revealing the atomic and electronic structures of the sample.



Procedure

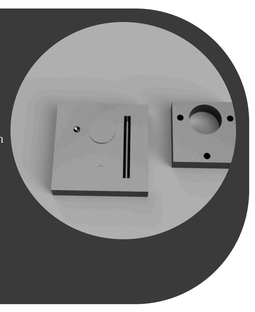
- Suspend the STM on a cloth hanging in the air to isolate it from mechanical vibrations.
- Start with screwing the upper screws until the tip seems sufficiently close to the sample. Then start using the lower screw to help close in on the sample until you are able to establish a current with the sample
- Once the connection has been established, run the STM scanner file by sending the serial command "ST"
- Once the data has compiled in the data scanner, run the post-processing code to add fourier transforms to see different patterns in the sample.

How is our design so cheap and portable?

- While designing the STM, we took a conventional STM and attempted to recreate the same mechanisms using cheaper designs. Although these mechanisms meant we lost considerable accuracy and resolution, they allow us to image atoms at a very cheap and affordable price.

The body

We decided to 3D print our frame for the structure since it allows us to go through multiple design iterations efficiently. Eventually, we plan to switch to metal which would reinforce the microscope's sturdiness



The Stepper

In order to bring the tip at a 1nm distance from the sample, we had to come up with a system that would allow the scanner head to travel at very small increments (nm). For this, we used a lever based system composed of 3 screws to get a reducing factor of 450x on a vertical distance of 3 µm giving us a step of 6nm per degree.

The piezo head

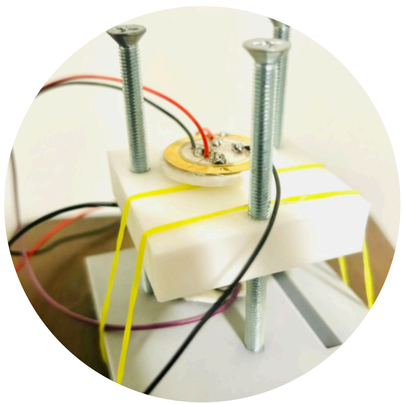
Because of the precise nature of atoms, we had to move at sub-nanometer distances in order to produce an accurate sample topography. To do this, we used a piezoelectric material (commonly found in buzzers) which gives 160nm / volt and ran it at very low voltages in the x and y direction.

Power system

To power our circuitry and the STM, we decided to use 2 rechargeable 9V batteries. These batteries can be easily transported and are more portable than the conventional power sources which tend to be bulky.

Conclusion

Our revolutionary prototype demonstrates the ability to produce images illustrating atomic structures within conductive materials, showcasing an unprecedented capability of elucidating atomic patterns in everyday items such as home utensils. Remarkably, this low cost device is able to obtain atomic lattice arrangements that match trends in metals like silver and gold, offering insights previously reserved for sophisticated laboratories. Utilizing advanced piezoelectric materials, we achieved an extraordinary x-y precision of 1 nm, allowing for minute exploration of sample surfaces. The dual-tier z-axis, including major-z precision of 6 nm for adjusting the sample tip distance with a screw lever system, and then a 1 nm minor precision to bring the tip close to the sample while scanning using piezo-material, ensures that the sample tip maintains optimal proximity for high-fidelity imaging. Our device is able to distinguish macroscopic topographical patterns in the sample, such as ravines or wave-like rise and fall patterns. We have meticulously engineered our system to minimize noise levels, reaching levels of around 0.04 to 0.06 for our setup. In terms of accessibility, we shattered conventional cost barriers, allowing for an assembly cost of just under $60 per unit for individual production, and $20 per unit for high volume manufacturing. Moreover, our microscope effectively runs on merely two 9-volt batteries, underscoring the emphasis our design has on combining high performance with sustainability and convenience. This not only paves the way for advanced research and quality control in various industries, but also allows hobbyist projects and educational environments to explore the nanoscopic world, fostering a greater appreciation for the microscopic world beyond our sight.



Results

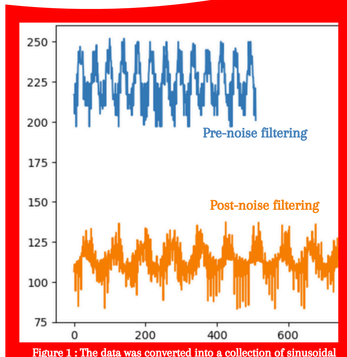


Figure 1 : The data was converted into a collection of sinusoidal waves of mixed frequencies, from which random noise was removed to highlight important features of the sample

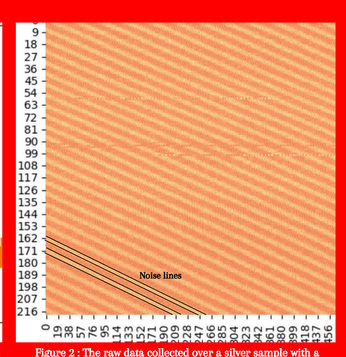


Figure 2 : The raw data collected over a silver sample with a resolution of a nanometer

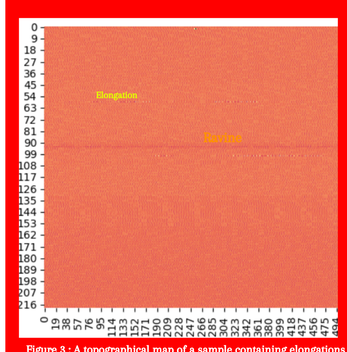


Figure 3 : A topographical map of a sample containing elongations and ravines on its surface.

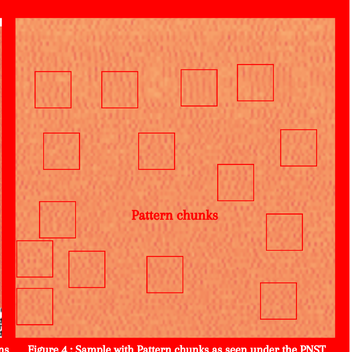


Figure 4 : Sample with Pattern chunks as seen under the PNST

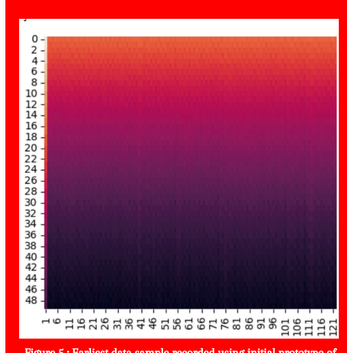


Figure 5 : Earliest data sample recorded using initial prototype of the PNST without any optimization and processing. In future models, the scanning tips and methods were greatly improved

Figure 6 : Early data recorded on the surface of a US penny (copper coating) which gave a result with distinct features but lacking the ability to clearly image atoms

Our PNST is able to identify distinct features in samples. Figure 1 shows the process by which we filtered noise from our samples, using Fourier transformations to increase the visibility of key features. Figure 2 shows a raw collection of data for a sample of silver, showing how the noise significantly obscures the features present in a sample. Figures 3 and 4 demonstrate some of the distinct features the PNST enables us to see, such as sudden elongations, pattern chunks, and ravines. Figure 9 and 10 show results from earlier models of the PNST. Figure 5 shows the earliest trial of the PNST, which resulted in a very obscured sample, and Figure 6 shows an improved sample with distinct features but ultimately unable to see individual atoms with clarity.

Materials

Material	Quantity	Cost
PLA plastic	40g	$1
Piezoelectric material	1	$0.5
Metal screws & nuts	3 pairs	$4
Conductive tip (Tungsten)	1	$1
Rechargeable 9V batteries	2	$9
Arduino	1	$20
Rubber bands	2	Negligible
Coin	1	Negligible
Circuitry	...	$20